

LAKSHMANAN R

B.E. Electronics & Communication Engineering | VLSI Design & Technology

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PROFESSIONAL SUMMARY

Third-year B.E. Electronics and Communication Engineering student specializing in VLSI Design and Technology at R.M.K. Engineering College (CGPA: 7.89). Winner of WE Hack at VIT Vellore with an Edge-AI rover. Hands-on experience in robotics, embedded systems, control systems (PID/ADRC), and EDA tooling, including a research internship at NIT Tiruchirappalli. Active IEEE member and quick learner who applies academic knowledge to real-world problems.

EDUCATION

B.E. Electronics and Communication Engineering (VLSI Design & Technology)	2024 – 2028
R.M.K. Engineering College, Chennai	CGPA: 7.89
Higher Secondary Certificate (12th Grade)	2024
Vijayantha Model School	88%
Secondary School Certificate (10th Grade)	
Vijayantha Model School	86%

TECHNICAL SKILLS

Programming Languages:	C, C++, Java, Verilog, SystemVerilog
Web Technologies:	HTML, JavaScript, React.js
Database:	MySQL
Hardware & EDA Tools:	PCB Designing, Cadence Virtuoso, Vivado (Xilinx), MATLAB
Embedded Platforms:	Raspberry Pi, Arduino, ESP32, Vega Aries V3
Protocols & Concepts:	VLSI Design Flow, Digital Logic, I2C, UART, ADC, Embedded Systems
Control & AI:	PID Control, ADRC, Edge AI, Machine Learning basics
Sensors & Hardware:	LiDAR (TF-Luna), Servo and DC Motors, Motor Drivers, Microcontrollers

EXPERIENCE

Research Intern – NIT Tiruchirappalli (ICE Department)	Summer 2026
- Worked on a 3-DOF robotic arm project with Active Disturbance Rejection Control (ADRC), under faculty supervision - Built PC-based servo control (GUI, serial communication) and closed-loop PID positioning with ADC feedback - Worked toward converting the PID controller into an ADRC-based controller for the arm	
Online Intern – CODSOFT	2024
- Studied industrial automation systems (PLC, HMI, batching plant, extrusion) in a manufacturing setting - Learned preventive maintenance workflows and industrial safety standards	

PROJECTS

Autonomous Edge-AI Environmental Inspection Rover (1st Prize, WE Hack)

- Raspberry Pi rover with on-device ML, TF-Luna LiDAR and environmental sensors

3-DOF Robotic Arm with PID / ADRC Control

- Servo-based arm (Vega Aries V3, PCA9685) with GUI control and closed-loop PID positioning

Verilog Analyzer + STA Timing + Power Estimator Tool (Offline Website)

- Browser-based offline tool for Verilog analysis, static timing analysis and power estimation

Study AI Platform

- AI-powered learning platform with quizzes, flashcards and progress tracking, built at Envision Hackathon

GitHub: github.com/priyadharshini-sip/study-ai

Line Tracer Robot

- Line-following robot with sensor-based path tracking and motor control

Obstacle Avoidance Bot (2nd Prize)

- Raspberry Pi and TF-Luna LiDAR robot for real-time obstacle detection and navigation

ACHIEVEMENTS, HACKATHONS & ACTIVITIES

WE Hack, VIT Vellore – 1st Prize

- Winner of the hackathon with a cash prize of INR 70,000 for the Autonomous Edge-AI Environmental Inspection Rover

Envision Hackathon 2024–2025

- Competed in Envision Hackathon, developing Study AI – an AI-powered student learning support platform

Conference – Dhanalakshmi Srinivasan

- Attended a conference at Dhanalakshmi Srinivasan

IEEE Student Member

- Active IEEE member focused on learning emerging technologies, participating in technical events, and innovating practical solutions

VIT Workshop

- Attended a workshop at VIT Chennai on how to use RHP Pearl

Volunteering

- NGO volunteer work (care and welfare activities)